DERIVATIVES QUIZ

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Name: Solutions

1. Use the **limit definition**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

to compute the derivative of

$$f(x) = \frac{1}{x}.$$

Check your answer using the Derivative Rules.

$$f'(x) = \lim_{h \to 0} \frac{\frac{1}{(x+h)} - \frac{1}{x}}{h} = \lim_{h \to 0} \frac{1}{h}\left(\frac{1}{(x+h)}\left(\frac{x}{x}\right) - \frac{1}{x}\frac{(x+h)}{(x+h)}\right)$$

$$= \lim_{h \to 0} \frac{x - (x+h)}{h \times (x+h)}$$

$$= \lim_{h \to 0} \frac{x - x - h}{h \times (x+h)}$$

$$= \lim_{h \to 0} \frac{-h}{h \times (x+h)}$$

$$= \lim_{h \to 0} \frac{-1}{x(x+h)}$$

$$= \frac{-1}{x(x+0)}$$

$$= \boxed{\frac{-1}{x^2}}$$

Check

Power Rule: $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx} x^{-1} = -1(x^{-2}) = \frac{-1}{x^2}.$

Quotient Rule: $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{\frac{d}{dx}(1)\, x - 1 \frac{d}{dx}(x)}{x^2} = \frac{0 \cdot x - 1(1)}{x^2} = \frac{-1}{x^2}.$

Date: February 15, 2019.